

Foundations of Artificial Intelligence

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Exercise Sheet 10

Due: May 10, 2026

Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

Exercise 10.1 (0.5+0.5+1+1+1+1+1 marks)

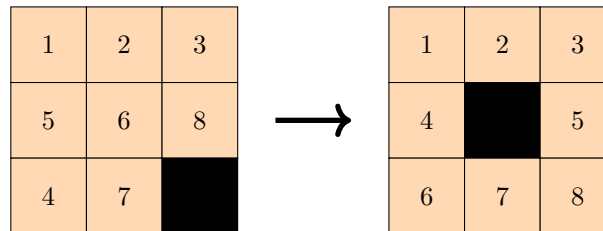
Consider the STRIPS planning task $\Pi = \langle V, I, G, A \rangle$ with $V = \{a, b, c, d, e, f\}$, $I = \{a, b\}$, $G = \{e, f\}$, and $A = \{a_1, a_2, a_3\}$ with $cost(a) = 1$ for all $a \in A$ and

$$\begin{array}{lll} pre(a_1) = \{a\} & add(a_1) = \{c\} & del(a_1) = \{e\} \\ pre(a_2) = \{b\} & add(a_2) = \{d\} & del(a_2) = \{c\} \\ pre(a_3) = \{c, d\} & add(a_3) = \{e, f\} & del(a_3) = \{\}. \end{array}$$

- (a) Provide $h^*(I)$, the cost of an optimal plan.
- (b) Provide $h^+(I)$, the cost of an optimal relaxed plan.
- (c) Provide the relaxed planning graph (RPG) for Π with enough layers for it to stabilize (i.e., up to a layer n such that $V_{n-1} = V_n$).
- (d) Compute $h^{max}(I)$. Provide the values for all nodes in the RPG.
- (e) Compute $h^{add}(I)$. Provide the values for all nodes in the RPG.
- (f) Compute $h^{FF}(I)$. Provide the marked RPG.

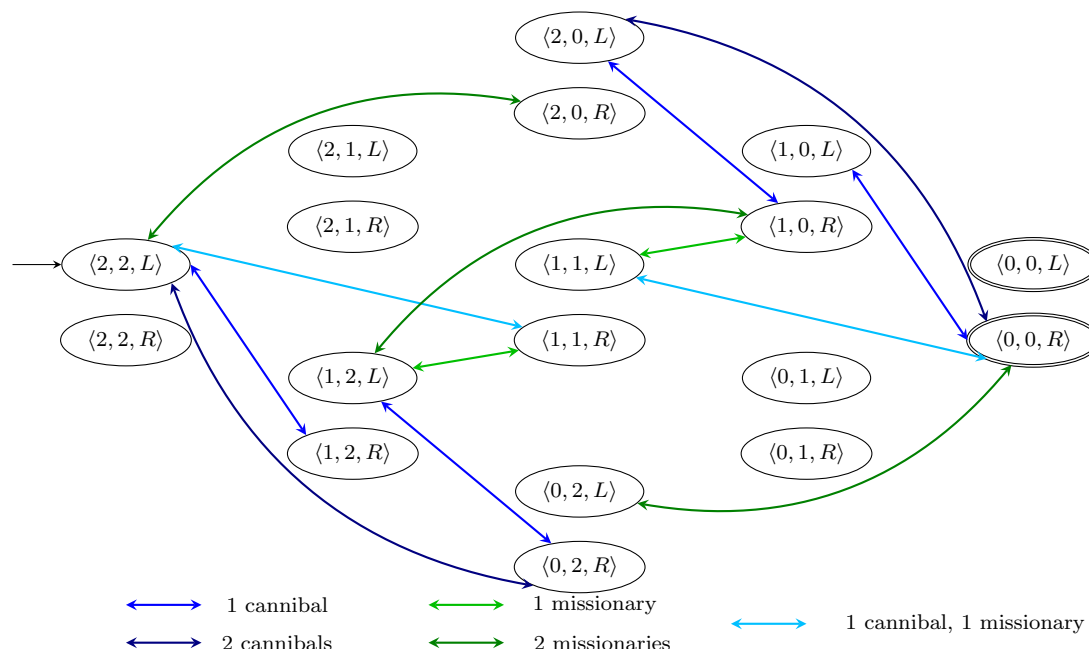
Exercise 10.2 (2 marks)

Formalize the following 8-puzzle instance in SAS⁺:



Exercise 10.3 (1+0.5+0.5+1 marks)

The following graph depicts the state space of a cannibals and missionary problem with two cannibals and two missionaries:



A state is represented as a tuple $\langle C, M, B \rangle$, where C and M (with $\text{dom}(C) = \text{dom}(M) = \{0, 1, 2\}$) denote how many cannibals and missionaries respectively are on the left shore, and B (with $\text{dom}(B) = \{L, R\}$) denotes on which shore the boat is. The cost of all operators is 1.

- Draw the abstract state space that is induced by the projection to $P = \{C, M\}$ in the same way as the above depiction of the concrete state space. (In particular, draw a new graph containing only abstract states instead of indicating in the one above which states are mapped to the same abstract state.)
- Use the abstraction from part (a) to derive a pattern database heuristic. Provide all database entries, i.e., the abstract distances for all states in the abstract state space.
- What is $h^{\{C, M\}}(I)$?
- For a cannibals and missionaries problem with n cannibals and n missionaries, what would $h^{\{C, M\}}(I)$ be? Justify your answer.

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in $\text{\LaTeX}$ are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.

- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
 - “We solved all exercises on our own.”
 - “We used ChatGPT to explain exercise 7 which we did not understand.”
 - “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)